Quantum Simulation of Spin-Dependent Charmonium Spectroscopy with Variational Quantum Algorithms

Master's Thesis

Lucas Froguel

2026

Contents

1 Introduction **1**

1.1 Quantum Computing . . . 1

1.1.1 Qubits and quantum states . . . 1

1.1.2 Quantum gates and circuits . . . 2

1.1.3 The NISQ era . . . 3

1.2 Variational Quantum Methods . . . 3

1.2.1 The variational principle . . . 3

1.2.2 The hybrid quantum–classical loop . . . 3

1.2.3 Ansatz design and practical considerations . . . 4

1.3 The Quarkonium Problem . . . 5

1.3.1 Heavy quark bound states . . . 5

1.3.2 The Cornell potential . . . 5

1.3.3 Spin-dependent interactions . . . 5

1.3.4 Quarkonium as a quantum simulation testbed . . . 7

1.4 Thesis Overview . . . 7

Bibliography **9**

Chapter 1

Introduction

The simulation of quantum systems lies at the heart of quantum computing's promise. As Feynman observed in 1982, classical computers face a fundamental barrier when modelling quantum mechanical systems: the Hilbert space of an n-particle system grows exponentially, rendering exact classical computation intractable for even modest system sizes [1]. Quantum computers, by contrast, naturally encode and manipulate quantum states, offering a path toward efficient simulation of physical systems that are otherwise inaccessible [2].

This thesis applies variational quantum algorithms to the spectroscopy of charmonium—bound states of a charm quark and its antiquark—including spin-dependent interactions that produce the experimentally observed mass splittings between states such as the η_c and J/ψ mesons. The quarkonium system is physically rich enough to probe the capabilities of near-term quantum algorithms across multiple quantum number channels and excited states, yet small enough to be tractable with only a few qubits.

The remainder of this chapter introduces the three pillars on which this work rests: the fundamentals of quantum computing (Section 1.1), variational quantum methods (Section 1.2), and the quarkonium problem (Section 1.3).

1.1 Quantum Computing

1.1.1 Qubits and quantum states

The fundamental unit of quantum information is the *qubit*, a two-level quantum system whose state is described by a vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{1.1}$$

where $\alpha, \beta \in \mathbb{C}$ are probability amplitudes and $\{|0\rangle\,, |1\rangle\}$ is the computational basis. Unlike a classical bit, which is deterministically 0 or 1, a qubit exists in a *superposition*

of both states. Measurement in the computational basis collapses the state, yielding $|0\rangle$ with probability $|\alpha|^2$ and $|1\rangle$ with probability $|\beta|^2$.

Any single-qubit pure state can be parameterized geometrically on the *Bloch sphere*:

$$|\psi\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|1\rangle\,, \tag{1.2}$$

with $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$.

The state of an n-qubit system lives in the tensor product space $(\mathbb{C}^2)^{\otimes n}$, spanned by 2^n computational basis states. States that cannot be written as a tensor product of individual qubit states are *entangled*—a uniquely quantum resource central to quantum speedups and with no classical analogue.

1.1.2 Quantum gates and circuits

Quantum computation proceeds through the application of *quantum gates*: unitary transformations acting on qubit registers. Single-qubit gates include the three Pauli operators,

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{1.3}$$

which satisfy $\sigma_i^2 = I$ and anticommute pairwise. These generate continuous rotation gates on the Bloch sphere,

$$R_\mu(\theta) = e^{-i\theta\,\mu/2} = \cos\frac{\theta}{2}\,I - i\sin\frac{\theta}{2}\,\mu, \qquad \mu \in \{X, Y, Z\}, \tag{1.4}$$

and the Hadamard gate $H = (X + Z)/\sqrt{2}$ creates equal superposition from a basis state.

Two-qubit entangling gates are essential for generating correlations. The controlled-NOT (CNOT) gate flips the target qubit conditioned on the control:

$$\mathrm{CNOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{1.5}$$

Arbitrary single-qubit rotations together with the CNOT form a *universal gate set*, capable of approximating any unitary transformation on n qubits to arbitrary precision [2].

A quantum algorithm is specified as a *quantum circuit*: a sequence of gates applied to a register of qubits, followed by measurement. Figure 1.1 illustrates a circuit that prepares a maximally entangled Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ from the initial state $|00\rangle$ using only a Hadamard gate and a CNOT.

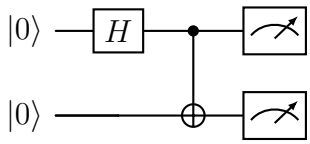

Figure 1.1: Quantum circuit preparing the Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. A Hadamard gate creates superposition on the first qubit; the CNOT entangles both qubits.

1.1.3 The NISQ era

Current quantum hardware operates in the *Noisy Intermediate-Scale Quantum* (NISQ) regime [3], characterized by processors with tens to hundreds of qubits subject to gate errors on the order of 10^{-3}–10^{-2}, limited qubit connectivity, and coherence times that restrict circuit depth. These constraints preclude the deep circuits required by fault-tolerant algorithms such as quantum phase estimation, motivating the development of shallow-depth methods tailored to near-term devices—chief among them the variational approach discussed in the next section.

1.2 Variational Quantum Methods

1.2.1 The variational principle

Variational methods offer a framework for extracting useful results from NISQ hardware. The central idea rests on the *Rayleigh–Ritz variational principle*: for a Hamiltonian H with ground state energy E_0,

$$E_0 \leq \langle\psi(\boldsymbol{\theta})| H |\psi(\boldsymbol{\theta})\rangle \equiv E(\boldsymbol{\theta}) \tag{1.6}$$

for any normalized trial state $|\psi(\boldsymbol{\theta})\rangle$ parameterized by classical variables $\boldsymbol{\theta} \in \mathbb{R}^p$. Minimizing $E(\boldsymbol{\theta})$ over the parameter space yields an upper bound on E_0, with equality when $|\psi(\boldsymbol{\theta})\rangle$ includes the true ground state in its span.

1.2.2 The hybrid quantum–classical loop

In the quantum computing implementation [4, 5], the trial state is prepared by a *parameterized quantum circuit* (also called an *ansatz*) applied to an initial reference state:

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta}) |0\rangle^{\otimes n}, \tag{1.7}$$

where $U(\boldsymbol{\theta})$ is a product of parameterized and fixed gates. The energy expectation value is then estimated on the quantum processor. For a Hamiltonian decomposed as a weighted

sum of Pauli strings,

$$H = \sum_{i=1}^{M} c_i \, P_i, \qquad P_i \in \{I, X, Y, Z\}^{\otimes n}, \tag{1.8}$$

each expectation value $\langle P_i \rangle$ is estimated independently through repeated state preparation and measurement in the appropriate Pauli basis. The total energy,

$$E(\boldsymbol{\theta}) = \sum_{i=1}^{M} c_i \, \langle P_i \rangle_{\boldsymbol{\theta}} \, , \tag{1.9}$$

is passed to a *classical optimizer* that updates $\boldsymbol{\theta}$ to minimize E. This loop repeats until convergence, as illustrated in Fig. 1.2.

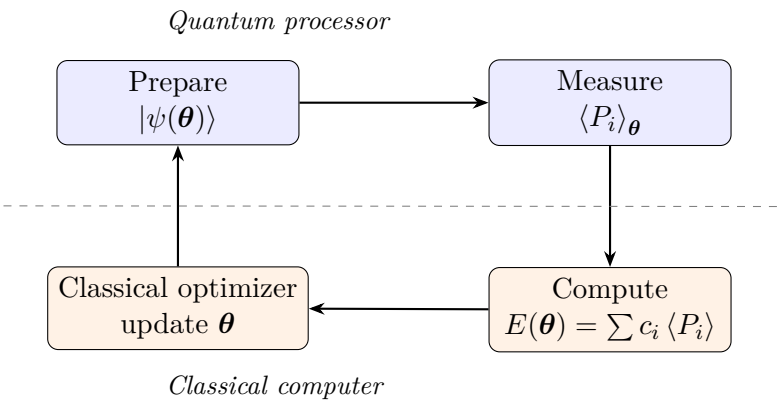

Figure 1.2: Schematic of the hybrid quantum–classical variational loop. The quantum processor prepares a parameterized state and estimates Pauli expectation values; the classical computer evaluates the energy and updates circuit parameters. The loop iterates until convergence.

1.2.3 Ansatz design and practical considerations

The choice of ansatz circuit is problem-dependent and critical to performance. *Hardware-efficient* ansatze use gates native to the target device, minimizing circuit depth at the cost of physical motivation. *Physically-motivated* ansatze, such as those inspired by unitary coupled cluster theory from quantum chemistry, encode domain knowledge into the circuit structure and may converge faster toward the physical ground state.

The hybrid approach offers several advantages for near-term hardware. The quantum processor handles the exponentially hard task of preparing and measuring many-body quantum states, while the classical optimizer navigates the energy landscape using well-established numerical methods. Circuit depths remain manageable, and variational methods exhibit a degree of inherent noise resilience: small perturbations in the energy surface due to hardware errors are partially absorbed by the optimization [5].

Beyond the ground state, variational frameworks extend naturally to excited states through orthogonality constraints and penalty methods, and to quantum dynamics through

variational formulations of time evolution. These extensions broaden the scope of problems addressable on near-term quantum hardware.

1.3 The Quarkonium Problem

1.3.1 Heavy quark bound states

Quantum chromodynamics (QCD), the gauge theory of the strong interaction, governs the binding of quarks and gluons into hadrons. Among the simplest hadronic systems are *quarkonia*: bound states of a heavy quark and its antiquark ($q\bar{q}$). Charmonium ($c\bar{c}$) systems have served as testing grounds for QCD since the simultaneous discovery of the J/ψ meson in 1974 at BNL [6] and SLAC [7]—an event known as the *November Revolution* in particle physics.

The large mass of the charm quark ($m_c \approx 1.27$ GeV$/c^2$) implies that the quark–antiquark pair moves non-relativistically inside the bound state, with typical velocities $v/c \sim 0.3$. This justifies a potential model approach [8] in which the dynamics are governed by the non-relativistic Schrödinger equation with an effective interquark potential.

1.3.2 The Cornell potential

The *Cornell potential* [8],

$$V(r) = -\frac{\kappa}{r} + \sigma\, r, \tag{1.10}$$

provides a successful phenomenological description of the $q\bar{q}$ interaction. It captures the two essential features of the strong force:

- **Short-range Coulomb attraction** ($-\kappa/r$), arising from one-gluon exchange between the color charges, analogous to the electromagnetic Coulomb potential but with QCD coupling;
- **Long-range linear confinement** ($\sigma\, r$), a consequence of the non-Abelian nature of QCD, modelled by a chromoelectric flux tube (string) with tension σ connecting the quark and antiquark.

Typical parameter values for charmonium are $\kappa \approx 0.52$ and $\sigma \approx 0.183$ GeV^2 [8]. Figure 1.3 shows the characteristic shape of this potential.

1.3.3 Spin-dependent interactions

The spin-independent Cornell potential alone does not account for the full charmonium spectrum. The mass splitting between states with different spin quantum numbers—for

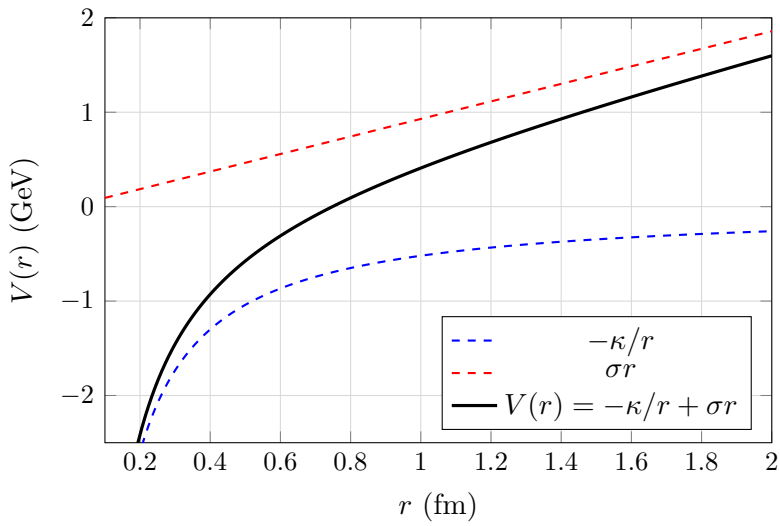

Figure 1.3: The Cornell potential as a function of interquark separation r. The dashed curves show the Coulomb ($-\kappa/r$, blue) and linear confinement (σr, red) components; the solid curve is their sum. At short distances the potential is dominated by one-gluon exchange; at large distances the linearly rising string tension ensures quark confinement.

instance, between the η_c ($^1S_0$, spin singlet) and the J/ψ ($^3S_1$, spin triplet)—requires spin-dependent corrections. At leading order, the dominant contribution for S-wave states is the hyperfine contact interaction [9],

$$V_{\text{hyp}}(r) = \frac{32\pi\,\alpha_s}{9\,m_c^2}\,\delta^{(3)}(\mathbf{r})\,\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}, \tag{1.11}$$

where α_s is the strong coupling constant and $\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}}$ is the spin–spin dot product. The eigenvalues of the latter operator distinguish singlet and triplet configurations:

$$\mathbf{S}_c \cdot \mathbf{S}_{\bar{c}} = \begin{cases} -\frac{3}{4} & \text{spin singlet } (S=0), \\ +\frac{1}{4} & \text{spin triplet } (S=1). \end{cases} \tag{1.12}$$

Quarkonium states are classified in spectroscopic notation $^{2S+1}L_J$, where S is the total spin, L the orbital angular momentum ($S, P, D, \ldots$ for $L = 0, 1, 2, \ldots$), and J the total angular momentum. The low-lying charmonium states relevant to this work are:

- $^1S_0$: spin singlet, S-wave $\longrightarrow$ η_c meson ($M \approx 2984$ MeV),
- $^3S_1$: spin triplet, S-wave $\longrightarrow$ J/ψ meson ($M \approx 3097$ MeV),
- $^1P_1$: spin singlet, P-wave $\longrightarrow$ h_c meson ($M \approx 3526$ MeV).

1.3.4 Quarkonium as a quantum simulation testbed

The full quarkonium Hamiltonian, including kinetic energy and both spin-independent and spin-dependent potentials, reads

$$H = -\frac{1}{2\mu}\nabla^2 + V(r) + V_{\text{hyp}}(r), \tag{1.13}$$

where $\mu = m_c/2$ is the reduced mass. Expanding the radial wavefunction in a truncated basis of N quantum harmonic oscillator (QHO) eigenstates yields a finite-dimensional matrix eigenvalue problem of size $N \times N$, whose entries are determined by the matrix elements of the kinetic and potential operators in the QHO basis.

This matrix can then be decomposed into a sum of Pauli operators (Eq. (1.8)) and encoded on a quantum register of $\lceil \log_2 N \rceil$ qubits, making it directly amenable to variational quantum algorithms. With $N = 3$–4 basis states, the problem requires only 2–3 qubits, placing it firmly within reach of current NISQ devices while retaining the essential physics: multiple spectroscopic channels, spin-dependent splittings, and a hierarchy of excited states.

This combination of physical richness and computational tractability makes quarkonium an ideal testbed for benchmarking quantum simulation algorithms against exact solutions and experimental data.

1.4 Thesis Overview

This thesis investigates the application of variational quantum algorithms to the charmonium spectroscopy problem described above. We implement and compare several algorithmic approaches for determining both ground and excited state energies across the $^1S_0$, $^3S_1$, and $^1P_1$ channels, assess their performance under realistic noise conditions, and evaluate error mitigation strategies. The goal is twofold: to demonstrate the feasibility of quantum simulation for hadronic physics on near-term hardware, and to provide a systematic comparison of methods that may inform future applications to more complex systems.

The thesis is organized as follows:

- **Chapter 2** develops the theoretical framework, including the Hamiltonian construction in the harmonic oscillator basis and the mapping to qubit operators.
- **Chapter 3** presents the quantum algorithms employed, covering both eigenvalue-based and time-evolution-based variational approaches.
- **Chapter 4** details the numerical results: energy spectra, convergence properties, and cross-method comparisons.

- **Chapter 5** addresses the effects of hardware noise and discusses error mitigation techniques.
- **Chapter 6** concludes with a summary of findings and an outlook on future directions.

Bibliography

[1] Richard P. Feynman. "Simulating Physics with Computers". In: *International Journal of Theoretical Physics* 21.6–7 (1982), pp. 467–488. DOI: 10.1007/BF02650179.

[2] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. 10th Anniversary. Cambridge University Press, 2010. ISBN: 978-1-107-00217-3.

[3] John Preskill. "Quantum Computing in the NISQ Era and Beyond". In: *Quantum* 2 (2018), p. 79. DOI: 10.22331/q-2018-08-06-79.

[4] Alberto Peruzzo et al. "A Variational Eigenvalue Solver on a Photonic Quantum Processor". In: *Nature Communications* 5 (2014), p. 4213. DOI: 10.1038/ncomms5213.

[5] Jarrod R. McClean et al. "The Theory of Variational Hybrid Quantum–Classical Algorithms". In: *New Journal of Physics* 18.2 (2016), p. 023023. DOI: 10.1088/1367-2630/18/2/023023.

[6] J. J. Aubert et al. "Experimental Observation of a Heavy Particle J". In: *Physical Review Letters* 33.23 (1974), pp. 1404–1406. DOI: 10.1103/PhysRevLett.33.1404.

[7] J.-E. Augustin et al. "Discovery of a Narrow Resonance in e^+e^- Annihilation". In: *Physical Review Letters* 33.23 (1974), pp. 1406–1408. DOI: 10.1103/PhysRevLett.33.1406.

[8] E. Eichten et al. "Charmonium: The Model". In: *Physical Review D* 17.11 (1978), pp. 3090–3117. DOI: 10.1103/PhysRevD.17.3090.

[9] N. Brambilla et al. "Heavy Quarkonium: Progress, Puzzles, and Opportunities". In: *The European Physical Journal C* 71 (2011), p. 1534. DOI: 10.1140/epjc/s10052-010-1534-9.